NEXAFORGE AI

Semantic Context Compression Pipeline

A methodology for reducing long-context cost while preserving task-critical meaning, evidence, and traceability

Document ID	Version	Owner	Classification
NF-AI-002	2.0	Context Systems Team	Confidential - Internal Use

Purpose

Specify how long conversations, retrieved documents, tool traces, and working notes are compressed into a smaller context without losing facts that are necessary for correct reasoning. The pipeline supports interactive assistants, agent workflows, document analysis, and long-running cases where naive truncation would cause inconsistency or hallucination.

Keywords: context compression, summarization, token optimization, long context, evidence preservation

Document Control

Field	Value
Status	Approved methodology baseline
Effective date	21 July 2026
Review cycle	Every 12 months or upon material change
Primary owner	Context Systems Team
Related standards	Security, privacy, software delivery, incident response, and data governance standards
Supersedes	Version 1.0 concise edition

How to Use This Document

Teams should use this document as a design and review guide, not as a substitute for engineering judgment. Sections describe mandatory controls, recommended practices, decision points, and evidence expected at release. Tailoring is permitted when documented and approved by the accountable owner. Any deviation that increases risk must include compensating controls and a review date.

1. Purpose and Objectives

Specify how long conversations, retrieved documents, tool traces, and working notes are compressed into a smaller context without losing facts that are necessary for correct reasoning. The pipeline supports interactive assistants, agent workflows, document analysis, and long-running cases where naive truncation would cause inconsistency or hallucination.

2. Scope and Applicability

This standard applies to new implementations, major changes, vendor replacements, and material expansions of systems that rely on semantic context compression pipeline. It covers prototypes that process real organizational data, pre-production pilots, and production services. Small experiments using synthetic data may follow a reduced process, but they must still document ownership, purpose, and data handling. The methodology does not replace legal, security, privacy, or domain-specific requirements; instead, it provides the engineering structure through which those requirements are implemented and verified.

3. Core Principles

Preserve decisions and evidence: Compression must retain commitments, user constraints, unresolved questions, source references, and data needed to reproduce conclusions.

Compress by function: Different content types require different treatment: instructions, facts, examples, logs, and conversational tone are not summarized in the same way.

Loss must be measurable: Every compression stage is evaluated against downstream tasks, not only linguistic similarity.

Originals remain addressable: Compressed representations store pointers to source spans so important details can be recovered on demand.

Recency is not importance: Recent content may be trivial, while older constraints may remain binding.

4. Lifecycle and Detailed Procedure

1. Context inventory

Classify input into system instructions, user preferences, task facts, source evidence, intermediate reasoning artifacts, tool outputs, and conversational filler. Assign retention priority and sensitivity tags.

- Required evidence: a reviewable artifact or trace showing that context inventory was completed.
- Decision record: assumptions, rejected alternatives, owner, and unresolved risks.
- Exit condition: measurable criteria are met before the workflow moves forward.

2. Structural segmentation

Split content along semantic and document boundaries rather than fixed token counts. Preserve headings, tables, code blocks, speaker turns, and source identifiers.

- Required evidence: a reviewable artifact or trace showing that structural segmentation was completed.
- Decision record: assumptions, rejected alternatives, owner, and unresolved risks.
- Exit condition: measurable criteria are met before the workflow moves forward.

3. Salience scoring

Score each segment for task relevance, obligation strength, novelty, evidence value, temporal validity, and future retrieval probability. Hard constraints and explicit decisions receive non-negotiable retention status.

- Required evidence: a reviewable artifact or trace showing that salience scoring was completed.
- Decision record: assumptions, rejected alternatives, owner, and unresolved risks.
- Exit condition: measurable criteria are met before the workflow moves forward.

4. Local compression

Generate compact summaries for segments using schemas appropriate to their function. Meeting-like text becomes decisions/actions/open issues; technical documents become definitions/requirements/exceptions; tool traces become inputs/results/errors.

- Required evidence: a reviewable artifact or trace showing that local compression was completed.
- Decision record: assumptions, rejected alternatives, owner, and unresolved risks.

- Exit condition: measurable criteria are met before the workflow moves forward.

5. Cross-segment consolidation

Merge duplicates, reconcile terminology, and surface contradictions. Contradictory facts are preserved as alternatives with provenance rather than collapsed into a single statement.

- Required evidence: a reviewable artifact or trace showing that cross-segment consolidation was completed.
- Decision record: assumptions, rejected alternatives, owner, and unresolved risks.
- Exit condition: measurable criteria are met before the workflow moves forward.

6. Evidence anchoring

Attach source pointers, identifiers, timestamps, or chunk references to compressed facts. For high-stakes facts, retain a short supporting excerpt or structured value rather than a paraphrase alone.

- Required evidence: a reviewable artifact or trace showing that evidence anchoring was completed.
- Decision record: assumptions, rejected alternatives, owner, and unresolved risks.
- Exit condition: measurable criteria are met before the workflow moves forward.

7. Budget allocation

Allocate the available token budget among instructions, current user turn, retained facts, evidence, and optional background. Reserve emergency budget for source expansion when verification requires it.

- Required evidence: a reviewable artifact or trace showing that budget allocation was completed.
- Decision record: assumptions, rejected alternatives, owner, and unresolved risks.
- Exit condition: measurable criteria are met before the workflow moves forward.

8. Downstream validation

Run target tasks against full and compressed contexts. Measure answer agreement, constraint retention, citation accuracy, and failure severity. Reject compression profiles that save tokens but degrade critical behavior.

- Required evidence: a reviewable artifact or trace showing that downstream validation was completed.
- Decision record: assumptions, rejected alternatives, owner, and unresolved risks.
- Exit condition: measurable criteria are met before the workflow moves forward.

5. Entry Criteria

Work may enter the methodology when a named owner, defined user outcome, initial data classification, and target environment are available. Teams should also identify downstream

consumers, irreversible actions, expected traffic, and the cost of wrong answers. If these inputs are unknown, the first deliverable is a discovery brief rather than an implementation.

6. Design Review Expectations

The design review should focus on concrete decisions and evidence. Reviewers inspect architecture diagrams, state or data schemas, model and tool dependencies, evaluation plans, privacy boundaries, fallback behavior, and operating metrics. Open issues must have owners and dates. A design is not approved merely because a model can demonstrate the happy path; the team must show how the system behaves with missing inputs, conflicting evidence, unavailable services, malformed outputs, and unauthorized requests.

7. Documentation and Traceability

Each release maintains a versioned record of requirements, decisions, prompts or policies, model identifiers, data or index versions, test results, known limitations, and approvers. Traceability should allow an incident responder to determine which configuration produced a specific output. Documentation must be concise enough to maintain but detailed enough for a new engineer to reproduce critical behavior.

8. Change Management

Changes are categorized as low, medium, or high impact. Low-impact changes include wording or monitoring adjustments with no behavior change. Medium-impact changes affect routing, prompts, thresholds, or non-sensitive data. High-impact changes include model replacement, new write capabilities, new sensitive data, changed jurisdictions, or altered decision authority. Medium and high-impact changes require targeted regression testing; high-impact changes repeat the relevant risk and approval reviews.

9. Operational Readiness

Before launch, the team confirms monitoring, alert ownership, on-call procedures, rollback, rate limits, access controls, data retention, incident communication, and user support. Synthetic probes should continuously test critical paths. Runbooks should describe both technical recovery and the user-facing behavior during degradation.

10. Continuous Improvement

Production evidence is reviewed on a regular cadence. Teams analyze failures by root cause rather than by surface symptom, distinguish model errors from retrieval, tool, data, policy, or interface errors, and prioritize improvements by user harm and frequency. Confirmed defects become regression tests. Metrics are segmented by use case, language, tenant, and risk tier to avoid hiding poor performance inside global averages.

12. Roles and Responsibilities

Role	Accountability
Context architect	Defines schemas, salience features, and source-linking strategy.
Application engineer	Integrates compression triggers and expansion calls.
Evaluation lead	Builds task-specific fidelity tests.
Privacy engineer	Defines retention and redaction rules.
Domain reviewer	Identifies facts whose omission would create material harm.

13. Measurement Framework

Metric	Definition and Use
Compression ratio	Original tokens divided by retained tokens.
Constraint retention	Binding constraints preserved after compression.
Fact fidelity	Correctly retained facts among facts required downstream.
Contradiction preservation	Conflicts represented explicitly rather than silently resolved.
Source recoverability	Compressed facts that can be mapped back to source spans.
Downstream quality delta	Performance difference between full and compressed context.
Expansion rate	Frequency with which the system must reopen original material.

Metrics must be segmented by use case and risk tier. Teams should define a baseline, target, alert threshold, and owner for each production metric. A metric without an action rule is considered observational rather than operational.

14. Worked Example

A customer-support case spans 120 messages, three contracts, and six tool calls. The compressor keeps the current issue, product identifiers, promised refund date, legal exception, and unresolved approval. Repeated greetings and duplicated troubleshooting steps are removed. Contract clauses are represented as structured obligations with page references. When the agent later drafts a response, it sees the binding promise and retrieves the exact clause before quoting it.

15. Risk Register and Controls

Risk	Primary Controls
Instruction loss	Pin system and policy instructions outside the lossy compression channel.
Semantic drift	Use structured summaries, source anchors,

	and periodic full-context regression tests.
Stale facts	Store timestamps and validity windows, then prefer newer verified facts when conflicts are temporal.
Over-compression of edge cases	Maintain exceptions and negative constraints as first-class fields.
Privacy leakage	Redact or tokenize sensitive details before summary generation and enforce retention schedules.
Summary accumulation	Periodically regenerate from authoritative source segments instead of repeatedly summarizing summaries.

16. Release Gate

- A named business and technical owner has approved the intended use and operating boundaries.
- Required architecture, security, privacy, and risk reviews are complete.
- Evaluation results meet minimum thresholds and no severe unresolved defect remains.
- Monitoring, alerting, rollback, and incident procedures have been tested.
- User-facing disclosures, approval checkpoints, and fallback behavior are implemented where required.
- All model, prompt, tool, data, index, and policy versions are recorded in the release manifest.

17. Review Checklist

- Is the user outcome specific, measurable, and appropriate for AI assistance?
- Are authority boundaries and prohibited actions explicit?
- Can every important output be traced to inputs, evidence, and configuration?
- Does the design handle missing, conflicting, stale, or malicious inputs?
- Are sensitive data, permissions, retention, and deletion controlled?
- Are severe failures represented in the evaluation suite?
- Can the service degrade or stop safely when dependencies fail?
- Are metrics connected to owners and response procedures?

18. Appendices

Context object schema

This appendix should be maintained as a project-specific artifact. It records the concrete implementation details, decisions, evidence, and approvals needed to apply the Semantic Context Compression Pipeline in a reproducible manner.

- Owner and review date

- Inputs and dependencies
- Decision criteria
- Required evidence
- Known limitations and follow-up actions

Salience scoring rubric

This appendix should be maintained as a project-specific artifact. It records the concrete implementation details, decisions, evidence, and approvals needed to apply the Semantic Context Compression Pipeline in a reproducible manner.

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Compression test set design

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Source expansion protocol

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- Decision criteria
- Required evidence
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Revision History

Version	Date	Summary
1.0	21 July 2026	Initial concise methodology edition.
2.0	21 July 2026	Expanded edition with

		detailed lifecycle, controls, roles, metrics, examples, risks, release gates, and appendices.